Eunoia Junior College
9729 H2 Chemistry
2017 Paper 3 Suggested Solution

1 (a) Hydrogen chloride, HCl, is stable to heat and does not decompose upon heating.

Hydrogen bromide, HBr, partially decompose upon heating:

$$2HBr(g) \xrightleftharpoons{\Delta} H_2(g) + \underset{\text{reddish-brown}}{Br_2(g)}$$

to give an equilibrium mixture of HBr, H_2 and reddish-brown gaseous Br_2.

Hydrogen iodide, HI, almost completely decompose upon heating:

$$2HI(g) \xrightarrow{\Delta} H_2(g) + \underset{\text{violet}}{I_2(g)}$$

to give H_2 and violet gaseous I_2.

The decreasing thermal stability of the hydrogen halide, HX, from HCl to HI is due to the increasing length and hence decreasing strength of the H–X bond, as the size of the halogen increases from Cl to I.

(b) (i) $MgCl_2$ has a **giant ionic structure** with **strong** electrostatic forces of attraction (**ionic bond**) between oppositely charged Mg^{2+} and Cl^- ions.

$SiCl_4$ has a **simple molecular structure** with **weak instantaneous dipole-induced dipole (id-id) attraction** between the **non-polar** $SiCl_4$ molecules.

Hence much more energy is required to pull apart the Mg^{2+} and Cl^- ions, than to pull apart the $SiCl_4$ molecules, as seen in the much higher melting point of $MgCl_2$ (714 °C) compared to that of $SiCl_4$ (–70 °C).

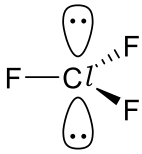
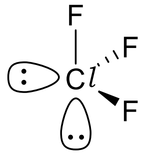
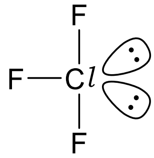
(b) (ii) Both SiF_4 and $SiCl_4$ have **simple molecular structures** with **weak id-id attraction** between the **non-polar** SiX_4 molecules.

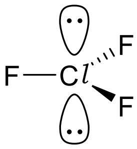
Since F is smaller than Cl, SiF_4 is smaller and has fewer electrons compared to $SiCl_4$. SiF_4 is thus **less polarisable**, leading to **weaker id-id attraction** and hence lower melting point compared to $SiCl_4$.

(b) (iii) Both MgF_2 and $MgCl_2$ have **giant ionic structures** with **strong** electrostatic forces of attraction (**ionic bond**) between oppositely charged ions.

The magnitude of the lattice energy $\propto \left|\dfrac{q^+q^-}{r_+ + r_-}\right|$ is a measure of the strength of the ionic bond. Since $r_{F^-} < r_{Cl^-}$, the magnitude of the lattice energy of MgF_2 will be higher than that of $MgCl_2$, implying that the ionic bond is stronger in MgF_2, leading to the higher melting point.

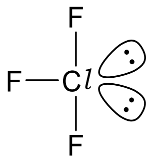
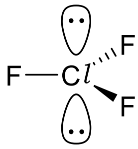
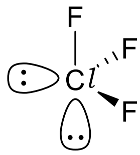
(Alternatively, since F^- is smaller than Cl^-, F^- has a **higher charge density**, and thus attracts the Mg^{2+} ion more strongly compared to Cl^-. As a result of the **stronger ionic bonds** in MgF_2, it has a higher melting point compared to $MgCl_2$.)

(c) (i) The three molecular arrangements are  ,  and  .

 would have no dipole moment.

(c) (ii) VSEPR theory assumes that electron pairs will take up positions where they are as far apart from one another as possible. In addition, orbitals containing lone pairs (lp) are larger than those containing bonded pairs (bp), and take up more space around the central atom, resulting in the repulsion between two lone pairs being more severe than that between a lone pair and bonded pair, which in turn is more severe than the repulsion between two bonded pairs.

The number of type of repulsion present in the three structures and hence the order of stability are:

	>		>	
1 × 120º lp-lp		1 × 180º lp-lp		1 × 90º lp-lp
4 × 90º lp-bp 2 × 120º lp-bp		6 × 90º lp-bp		3 × 90º lp-bp 2 × 120º lp-bp 1 × 180º lp-bp
2 × 90º bp-bp 1 × 180º bp-bp		3 × 120º bp-bp		2 × 90º bp-bp 1 × 120º bp-bp

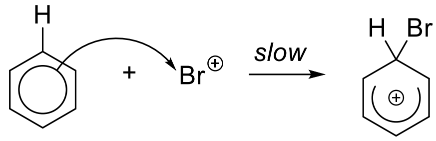
The least stable arrangement is where the two lone pair electrons are at 90º to each other. The most stable arrangement is where both lone pairs of electrons are in the equatorial position, which minimises the number of 90º lone pair-bond pair repulsion.

(d) (i) The product is bromobenzene. This is because bromine is less electronegative compared to chlorine and hence the bond in BrC*l* will be polarised as $\overset{\delta+}{\mathrm{Br}}—\overset{\delta-}{\mathrm{C}l}$, leading to formation of $Br^{+}AlCl_4^{-}$ with the Lewis acid, where the electrophilic Br^+ attacks the nucleophilic benzene ring.

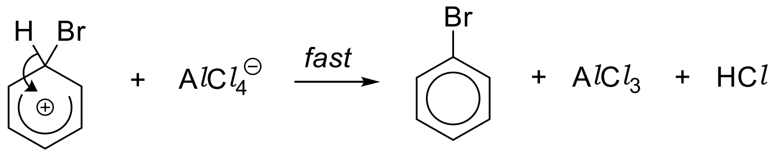
(d) (ii) **Step 1:** Formation of strong electrophile Br^+

$$\overset{\delta+}{Br}-\overset{\delta-}{Cl} + AlCl_3 \xrightarrow{fast} \overset{\delta+}{Br}-Cl \rightarrow AlCl_3 \longrightarrow Br^{\oplus}AlCl_4^{\ominus}$$

Step 2: Electrophilic attack of Br^+ on the π electron-rich benzene ring



Step 3: Regeneration of benzene ring and Lewis acid catalyst



(e) (i) $K_{sp}(AgCl) = [Ag^+][Cl^-]$ mol^2 dm^{-6}

(e) (ii) There are two sources of Ag^+ (aq), namely, from the $AgNO_3$ and dissolved AgC*l*. So $[Ag^+] = 0.50 + [Cl^-]$.

Substituting into the K_{sp} expression:

$$K_{sp}(AgCl) = [Ag^+][Cl^-] = (0.50 + [Cl^-])[Cl^-]$$

Since $[Cl^-]$ is very small, $0.50 + [Cl^-] \approx 0.50$. Hence

$$K_{sp}(AgCl) \approx (0.50)[Cl^-]$$

$$[Cl^-] = \frac{K_{sp}(AgCl)}{0.50} = \underline{\mathbf{4.00 \times 10^{-10}\ mol\,dm^{-3}}}$$

(e) (iii) When NH_3(aq) is added, Ag^+ reacts with NH_3 :

$$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq)$$

leading to a decrease in the concentration of Ag^+(aq). As a result, the equilibrium

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

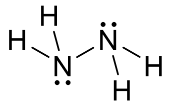
shifts to the right in attempt to counter this decrease, resulting in an **increase** in the solubility of AgC*l*.

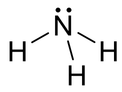
When NaC*l*(aq) is added, there is an increase in the concentration of Cl^-(aq). As a result, the equilibrium

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

shifts to the left in attempt to counter this increase, resulting in a **decrease** in the solubility of AgC*l*.

2 (a) (i) On the average, each hydrazine molecule can form 2 hydrogen bonds per molecule as compared to the ammonia molecule which can only form 1 hydrogen bond per molecule, that is, the hydrogen bonding in hydrazine is more extensive.


hydrazine

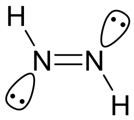

ammonia

In addition, hydrazine having almost twice as many electrons as ammonia will be more polarisable and hence exert stronger instantaneous dipole-induced dipole attraction.

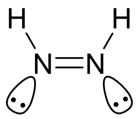
(a) (ii) The equation between hydrazine and sulfuric acid is

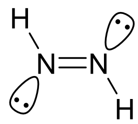
$$N_2H_4 \ + \ H_2SO_4 \ \rightarrow \ N_2H_5^+ \ + \ HSO_4^-$$

$N_2H_5^+$ is the conjugate acid of the base N_2H_4, while HSO_4^- is the conjugate base of the acid H_2SO_4.

(b) The structure of diazene is  , with a H–N–N bond angle **slightly less than** the ideal trigonal planar angle of **120°**, possibly **118°**.

Each N of the N=N double bond **possesses one lone pair and one hydrogen atom**. Hence, two non-superimposable isomers can arise due to **restricted rotation about the N=N double bond** as the two hydrogen can be on the same side or opposite side of the N=N:


cis-diazene


trans-diazene

(c) The particles in the gaseous mixture formed when solid **A** sublimes are NH_3 and HN_3.

The particles in a solution of **A** in water are NH_4^+ and N_3^-.

(This is similar to the familiar reaction between NH_3 and HCl to give solid $NH_4^+Cl^-$, which sublimes and dissociates back into NH_3 and HCl upon heating)

(d) (i) $5NaN_3(s) + NaNO_3(s) \rightarrow 3Na_2O(s) + 8N_2(g)$

(d) (ii) $n_{NaN_3} = \dfrac{400 \text{ g}}{(23.0 + 3 \times 14.0) \text{ g mol}^{-1}} = 6.154 \text{ mol}$

$n_{N_2} \text{ produced} = \dfrac{8}{5} \times n_{NaN_3} = \dfrac{8}{5} \times 6.154 = 9.846 \text{ mol}$

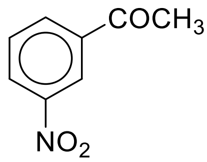
Using the ideal gas equation,

$$pV = nRT$$

$$p(100 \times 10^{-3}) = 9.846 \times 8.31 \times 298$$

$$p = 243828 \text{ Pa} \approx \underline{\mathbf{2.44\,kPa}}$$

(e) (i) Amide

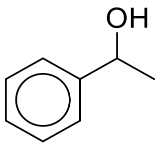
(e) (ii) **C** is  .

Reagent: a mixture of **concentrated HNO_3 and concentrated H_2SO_4**
Condition: maintain the temperature at **55–60 ºC**.

(e) (iii) Step 3
Reagent: **C_2H_5Cl with $AlCl_3$**
Condition: **Room temperature**

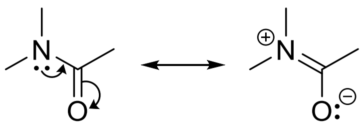
Step 4
Reagent: **Cl_2, excess ethylbenzene**
Condition: **UV light/heat**

(e) (iv) Reagent: **Aqueous NaOH**
Condition: **Reflux**

E is 

(e) (v) Reagent: **Aqueous NaOH**
Condition: **Reflux**

(e) (vi) **D** is essentially neutral while **F** will be slightly basic. In order words, **D** is less basic than **F.** Although the lone pair of electrons on N in **F** is delocalised into the benzene ring and hence less available for donation to an acid compared to an aliphatic amine. However, this delocalisation is less effective than the delocalisation of the lone pair on N in amide **D** into the carbonyl group:



as oxygen being more electronegative is better at accommodating the pair of electrons from N compared to the benzene ring. As a result, the lone pair on N in **D** is essentially not available for donation to an acid, making **D** less basic than **F**.

3 (a) Copper is able to exhibit variable oxidation states (0, +1, +2, +3) in its compounds, while calcium only forms compounds in the +2 oxidation state. This is because copper is able to use variable number of its 4s and 3d electrons, which are very close in energy, when forming compounds, while calcium invariably losses its two valence 4s electrons as electrons in the penultimate 3p orbitals are much lower in energy.

Copper has a higher melting point compared to calcium. This is because copper able to use both its 4s and 3d electrons while calcium only uses its two valence 4s electrons for metallic bonding, hence copper forms stronger metallic bonds, leading to the higher melting point.

(b) (i) In the presence of the 6 H_2O ligands, the five 3d orbitals in the central metal ion is split into two sets of different energy. Electron(s) in the lower energy orbitals can absorb visible light of a particular wavelength (*i.e.* energy) and get promoted to the higher energy orbitals (d-d transition). The colour is due to this absorption of light and the colour observed is the complement of the colour absorbed.

Cu^{2+} has an electronic configuration of [Ar] $3d^9$, while Zn^{2+} has an electronic configuration of [Ar] $3d^{10}$.

	Cu^{2+}			Zn^{2+}	
	↑↓ $d_{x^2-y^2}$	↑ d_{z^2}		↑↓ $d_{x^2-y^2}$	↑↓ d_{z^2}
↑↓ d_{xy}	↑↓ d_{xz}	↑↓ d_{yz}	↑↓ d_{xy}	↑↓ d_{xz}	↑↓ d_{yz}

In the case of Cu^{2+}, there is a singly occupied orbital in the higher energy set, hence allowing electrons in the lower energy set to be promoted and appear coloured. However, in Zn^{2+} all the five 3d orbitals are fully filled and d-d transition is not possible, hence appearing colourless. (Note that electrons in the 3d orbitals can be promoted to the much higher energy 4s orbitals. In this case, the energy falls outside of the visible region, into the ultraviolet region of the electromagnetic spectrum, and hence appears colourless)

(b) (ii) A complex is formed when ligands donates a pair of electrons to a metal ion. Cu^{2+}(aq) contains the complex $\left[Cu(H_2O)_6\right]^{2+}$, formed when 6 water ligands each donates a pair of electrons to the central Cu^{2+} ion. Upon reaction with ammonia, four of the water ligands in $\left[Cu(H_2O)_6\right]^{2+}$ will be substituted by NH_3 ligands, to give a new complex $\left[Cu(NH_3)_4(H_2O)_2\right]^{2+}$:

$$\left[Cu(H_2O)_6\right]^{2+} + 4NH_3 \rightleftharpoons \left[Cu(NH_3)_4(H_2O)_2\right]^{2+} + 4H_2O$$

(c) n_{KMnO_4} reacted with $\mathbf{G} = \frac{16.4}{1000} \times 2.00 \times 10^{-3} = 3.28 \times 10^{-5}$ mol

Since $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

n_{e^-} lost by $\mathbf{G} = 5 \times 3.28 \times 10^{-5} = 1.64 \times 10^{-4}$ mol

Since **G** is oxidised to VO_2^+, where vanadium is in the +5 oxidation state, **G** must contain vanadium in the +2 (V^{2+}), +3 (V^{3+}) or +4 (VO^{2+}) oxidation state.

Since $E^\ominus\left(Zn^{2+}\middle|Zn\right) = -0.76$ V is more negative that of $E^\ominus\left(VO^{2+}\middle|V^{3+}\right) = +0.34$ V and $E^\ominus\left(V^{3+}\middle|V^{2+}\right) = -0.26$ V, **G** must have been reduced to the V^{2+}(aq) ion, *i.e.* the oxidation state of vanadium in **H** is **<u>+2</u>**.

Oxidation of **H** to VO_2^+: $V^{2+} + 2H_2O \rightarrow VO_2^+ + 4H^+ + 3e^-$

n_{KMnO_4} reacted with $\mathbf{H} = \frac{24.6}{1000} \times 2.00 \times 10^{-3} = 4.92 \times 10^{-5}$ mol

n_{e^-} lost by $\mathbf{H} = n_{e^-}$ gained by $MnO_4^- = 5 \times 4.92 \times 10^{-5} = 2.46 \times 10^{-4}$ mol

$n_{\mathbf{H}} = n_{V^{2+}} = \frac{2.46 \times 10^{-4}}{3} = 8.20 \times 10^{-5}$ mol $= n_{\mathbf{G}}$

Let **G** contains vanadium in the +*n* oxidation state $(2 \le n \le 4)$. Each mole of **G** will lose (5–*n*) moles of electrons when oxidised to VO_2^+: $V^{n+} + (5-n)e^- \rightarrow V^{5+}$

Hence, $5 - n = \frac{n_{e^-} \text{ lost by } \mathbf{G}}{n_{\mathbf{G}}} = \frac{n_{e^-} \text{ lost by } \mathbf{G}}{n_{\mathbf{H}}} = \frac{1.64 \times 10^{-4}}{8.20 \times 10^{-5}} = 2$

So, $n = 5 - 2 = 3$

Thus, the oxidation state of vanadium in **G** is **<u>+3</u>**.

<u>Alternative solution</u>

Assuming we have *x* moles of **G** in 25 cm^3, $x(5-n) = 1.64 \times 10^{-4}$

n_{KMnO_4} reacted with $\mathbf{H} = \frac{24.6}{1000} \times 2.00 \times 10^{-3} = 4.92 \times 10^{-5}$ mol

n_{e^-} lost by $\mathbf{H} = 5 \times 4.92 \times 10^{-5} = 2.46 \times 10^{-4}$ mol

Let **H** contains V in the +*m* oxidation state (*m* < *n* since **G** is reduced by Zn). Each mole of **H** will lose (5–*m*) moles of electrons when oxidised to VO_2^+

Hence, $x(5-m) = 2.46 \times 10^{-4}$.

Taking ratio of the two expression,

$$\frac{x(5-n)}{x(5-m)} = \frac{1.64 \times 10^{-4}}{2.46 \times 10^{-4}}$$

$$\frac{5-n}{5-m} = \frac{2}{3}$$

$$15 - 3n = 10 - 2m$$

$$3n - 2m = 5$$

The only possible integer solution with *n* < 5 is *n* = 3 and *m* = 2. Hence the oxidation state in solution **G** is +3 and in solution **H** is +2.

$x = \frac{1.64 \times 10^{-4}}{2} = \frac{2.46 \times 10^{-4}}{3} = 8.20 \times 10^{-5}$.

Hence, $[V^{3+}]$ in $\mathbf{G} = \frac{8.20 \times 10^{-5} \text{ mol}}{\frac{25.0}{1000} \text{ dm}^3} = \underline{\mathbf{3.28 \times 10^{-3} \text{ mol dm}^{-3}}}$

(d) **I** decolourises bromine water ⇒ **I** undergoes electrophilic addition ⇒ presence of C=C.

Comparing $C_6H_8O_2$ with C_6H_{14}, degree of unsaturation $= \frac{14-8}{2} = 3$. Likely that **I** contains C=C and/or C=O and/or a ring.

I does not react with sodium metal ⇒ no redox ⇒ absence of –OH.

I does not react with alkaline aqueous iodine ⇒ absence of $–CH(OH)CH_3$ and $–COCH_3$ group.

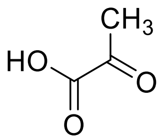
Solution of **I** is optically active ⇒ **I** contains a chiral centre.

$\underset{\mathbf{I}}{C_6H_8O_2} + H_2O \xrightarrow{H_2SO_4(aq)} \underset{\mathbf{J}}{C_6H_{10}O_3}$: **I** undergoes hydrolysis when heated with H_2SO_4(aq) ⇒ **I** contains a cyclic ester (lactone) group and **J** is a hydroxyacid.

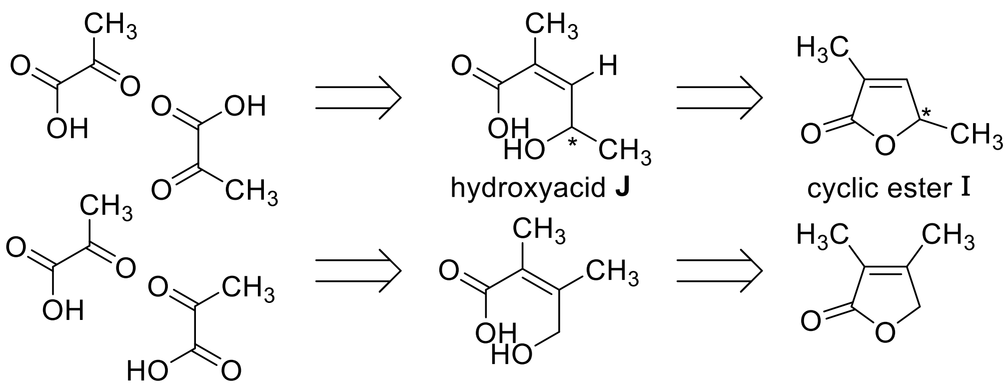
$\underset{\mathbf{J}}{C_6H_{10}O_3} + 4[O] \xrightarrow[\Delta]{KMnO_4} 2\underset{\mathbf{K}}{C_3H_4O_3} + H_2O$: C=C double bond in **I** undergoes oxidative cleavage.

K reacts with sodium metal ⇒ redox reaction ⇒ presence of $–CO_2H$.

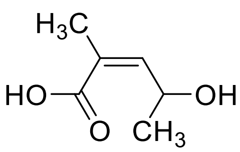
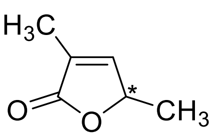
K reacts with alkaline aqueous iodine ⇒ oxidative cleavage ⇒ presence of –$COCH_3$.

K is 

Since **K** is formed when the C=C bond in hydroxyacid **J** undergoes oxidative cleavage, there are two ways to piece **K** to form **J** and then **I**:



Only difference is that only one will have a chiral centre, leading to optical activity.

Hence, **J** is , and **I** is 

When heated with H_2SO_4(aq), the ester (latone) group is hydrolysed to afford the hydroxyacid **J**, which then undergoes oxidative cleavage with concomitant oxidation of the methyl carbinol fragment, when treated with acidified $KMnO_4$ to give two identical products, the methyl ketoacid **K**.

4 (a) (i) The oxidation state of each of the carbon is $\overset{-3}{C}H_3\overset{-2}{C}H_2\overset{-2}{C}H_2\overset{+3}{C}O_2H$. Hence the average oxidation number of carbon is $\frac{-3-2-2+3}{4} = \mathbf{-1}$.

(a) (ii) Assuming $CH_3CH_2CH_2CO_2H + (2m-2)H_2O \rightarrow nCH_4 + mCO_2$, so $n + m = 4$ ..... (1)

Oxidation state of C in CH_4 is –4 and in CO_2 is +4, so $-4 = n(-4) + m(+4)$ ..... (2)

4(1) + (2): $8m = 12 \Rightarrow m = \frac{3}{2}$. Substituting into (1): $n = \frac{5}{2}$.

Hence, equation is $CH_3CH_2CH_2CO_2H + H_2O \rightarrow \frac{5}{2}CH_4 + \frac{3}{2}CO_2$

or

Since the oxidation number of carbon decreases by 3 units upon reduction to CH_4 (–1 in $CH_3CH_2CH_2CO_2H$ to –4 in CH_4), and increases by 5 units upon oxidation to CO_2 (–1 in $CH_3CH_2CH_2CO_2H$ to +4 in CO_2), the mole ratio of CH_4 to CO_2 must be 5 : 3. Hence, $2CH_3CH_2CH_2CO_2H \rightarrow 5CH_4 + 3CO_2$. Balancing the number of H's and O's with H_2O (since their oxidation numbers do not change) gives

$$2CH_3CH_2CH_2CO_2H + 2H_2O \rightarrow 5CH_4 + 3CO_2$$

or

[O] : $CH_3CH_2CH_2CO_2H + 6H_2O \rightarrow 4CO_2 + 20H^+ + 20e^-$

[H] : $CH_3CH_2CH_2CO_2H + 12H^+ + 12e^- \rightarrow 4CH_4 + 2H_2O$

[O]×3 + [H]×5 : $8CH_3CH_2CH_2CO_2H + 8H_2O \rightarrow 20CH_4 + 12CO_2$ or

$$2CH_3CH_2CH_2CO_2H + 2H_2O \rightarrow 5CH_4 + 3CO_2$$

(a) (iii) Passing the gaseous product mixture through an aqueous solution of NaOH.

(a) (iv) Pressure of remaining $CH_4 = \frac{5}{8} \times 1.5 \times 10^5 = 93750\ \text{Pa} \approx \underline{\mathbf{93.8\ kPa}}$

(b) (i) Using $\Delta H_r^\ominus = \sum \Delta H_f^\ominus(\text{products}) - \sum \Delta H_f^\ominus(\text{reactants})$,

$$\Delta H_r^\ominus = \frac{5}{2}\Delta H_f^\ominus(CH_4) + \frac{3}{2}\Delta H_f^\ominus(CO_2) - \Delta H_f^\ominus(CH_3CH_2CH_2CO_2H) - \Delta H_f^\ominus(H_2O)$$

$$= \frac{5}{2}(-75) + \frac{3}{2}(-394) - (-534) - (-286) = \underline{\mathbf{+41.5\ kJ\ mol^{-1}}}$$

(b) (ii) Using $\Delta G_r^\ominus = \Delta H_r^\ominus - T\Delta S_r^\ominus$,

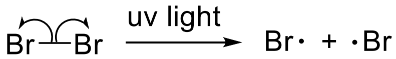
$$-207 = +41.5 - 298\Delta S_r^\ominus$$

$$\Delta S_r^\ominus = +0.834\ \text{kJ mol}^{-1}\ \text{K}^{-1} = \underline{\mathbf{+834\ J\ mol^{-1}\ K^{-1}}}$$

The sign of $\Delta S_r^\ominus$ is positive, meaning there is an increase in the entropy of the system which can be rationalised based on the fact that the reaction leads to the formation of 4 moles of gaseous particles per mole of aqueous butanoic acid and water reacted, resulting in a large increase in the disorderness of the system.

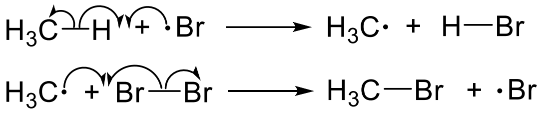
(c) (i) The mechanism of the reaction between methane and bromine is

initiation :



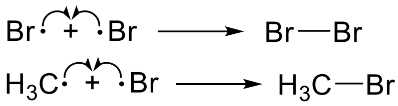
Formation of radicals by homolytic bond cleavage (this generally requires heat or light)

propagation :



Reaction of a radical to product a *new* product radical

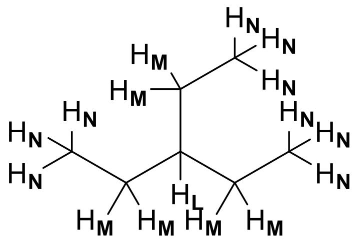
termination :

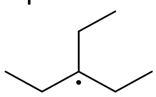
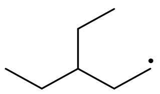


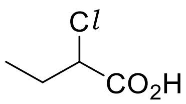
Coupling of two radicals to form only non-radical products

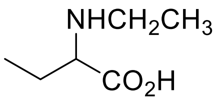
(c) (ii) The Br–Br bond has a bond dissociation energy of 193 kJ mol^{-1}, compared to 244 kJ mol^{-1} for the dissociation of the C*l*–C*l* bond. Less energy is required to break the Br–Br bond homolytically and hence longer wavelength yellow light is adequate, while higher energy blue or ultraviolet light is needed to cleave the stronger C*l*–C*l* bond.

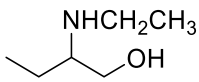
(c) (iii) Based on the number of each type of hydrogen present in 3-ethylpentane, the relative proportions of **L : M** : **N** is likely to be **<u>1 : 6 : 9</u>**.



(c) (iv) Despite there being 9 times more H_N compared to H_L, the radical formed when H_L is abstract in the first propagation step, , is a tertiary radical, while the corresponding radical formed when H_N is abstracted, , is a primary radial. Since alkyl groups exert electron donating inductive effect, the tertiary radical is much more stable and thus formed much more quickly, compared to the primary radical. In fact, the tertiary radical is formed about 18 times more quickly than the primary radical, resulting in the 2 : 1 ratio of **L** : **N** observed.

(d) **Q** is 

, **R** is 

and **S** is 

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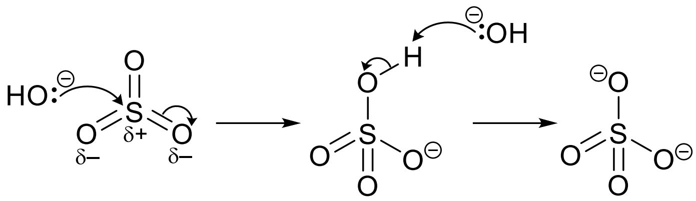
5 **(a)** Moving from sodium to sulfur of Period 3, the oxides changes from basic oxides (*e.g.* Na_2O) to amphoteric oxide (Al_2O_3) and finally to acidic oxides (*e.g.* SO_2).

Metallic oxides such as sodium oxide, Na_2O, are basic due to presence of the O^{2-} anion. Na_2O reacts with aqueous acids, *e.g.* hydrochloric acid, to give the salt, NaC*l*, and water:

$$\mathrm{xNa_2O(s) + 2HC}l\mathrm{(aq) \rightarrow 2NaC}l\mathrm{(aq) + H_2O(}l\mathrm{)}$$

On the other hand, covalent oxides such as sulfur trioxide, SO_3, are acidic due to the presence of $\overset{\delta+}{X}=\overset{\delta-}{O}$ double bond with an electron-deficient heteroatom X (X = P or S). SO_3 reacts with aqueous alkali, *e.g.* sodium hydroxide, to give $NaHSO_4$, and eventually the salt, Na_2SO_4, and water:



$$SO_3(g) + NaOH(aq) \rightarrow NaHSO_4(aq)$$
$$NaHSO_4(aq) + NaOH(aq) \rightarrow Na_2SO_4(aq) + H_2O(l)$$

Oxides such as aluminium oxide, Al_2O_3, which is ionic with covalent character will exhibit both characteristics, being amphoteric in nature, reacting with both aqueous acid and alkali. With hydrochloric acid, it gives the salt $AlCl_3$ and water:

$$Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$$

With sodium hydroxide, it reacts to give the salt sodium aluminate, $NaAlO_2$:

$$Al_2O_3(s) + 2NaOH(aq) \rightarrow 2NaAlO_2(aq) + H_2O(l)$$

(b) (i) The oxidation state of Cl increases by **2 units** from +5 in $KClO_3$ to +7 in $KClO_4$, and decreases by **1 unit** from +5 in $KClO_3$ to +4 in ClO_2.

Hence, for every $KClO_3$ oxidised to $KClO_4$, $2KClO_3$ will be reduced to $2ClO_2$, that is, $3KClO_3$ will give $KClO_4$ and $2ClO_2$:

$$3KClO_3 + 2H_2SO_4 \rightarrow KClO_4 + 2ClO_2 + 2KHSO_4 + H_2O$$

(b) (ii) The energy cycle is

$$\begin{tikzcd}
 & \text{2 O=C}l\text{=O(g)} \arrow[dl, "\text{4B.E.(C}l\text{=O)}"'] \arrow[dr, "-204\text{ kJ}"] & \\
\text{2C}l\text{(g) + 4O(g)} & & Cl_2\text{(g) + 2}O_2\text{(g)} \arrow[ll, "\text{B.E.(C}l\text{–C}l\text{) + 2B.E.(O=O)}"]
\end{tikzcd}$$

By Hess's Law,

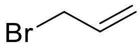
$$\begin{aligned}
4\text{B.E.}(Cl{=}O) &= -204 + \text{B.E.}(Cl-Cl) + 2\text{B.E.}(O{=}O) \\
&= -204 + 244 + (2 \times 496) \\
&= +1032 \text{ kJ} \\
\text{B.E.}(Cl{=}O) &= \underline{\mathbf{+258\ kJ\ mol^{-1}}}
\end{aligned}$$

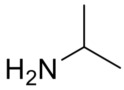
(b) (iii) Since the decomposition of 2 moles of $ClO_2(g)$ leads to the formation of 3 moles of gaseous products, there is an increase in the entropy, ΔS.

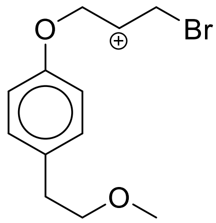
Using $\Delta G = \Delta H - T\Delta S$, since ΔH is negative, ΔS is positive and T is always positive, ΔG will be more negative than ΔH. In other words, the sign of ΔG like ΔH will be negative, while the numerical value of ΔG will be larger than that of ΔH.

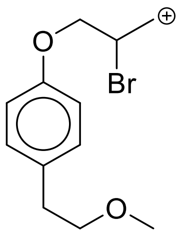
(c) (i) Step 3 is a **reduction**, while step 4 is an **acid-base neutralisation** reaction.

(c) (ii) The mechanisms are
Step 1: Electrophilic substitution
Step 2: Nucleophilic substitution
Step 6: Electrophilic addition

(c) (iii) Reagent **T** in step 5 is .

(c) (iv) Reagent **U** in step 7 is .

(c) (v) Step 6 involves electrophilic attack of the Br_2 at the C=C double bond to give an intermediate carbocation. If Br attaches to the terminal end of the C=C double bond, a secondary carbocation, 

, is formed. On the other hand, if Br attaches to the internal C of the C=C, a less stable primary carbocation intermediate, 

, is formed. The former carbocation being more stable will be formed in preference and hence subsequently a nucleophilic water molecule attacks the carbocation to give the bromohydrin isomer shown.